IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-37

Phase Shifted Full Bridge (PSFB) Converter - I

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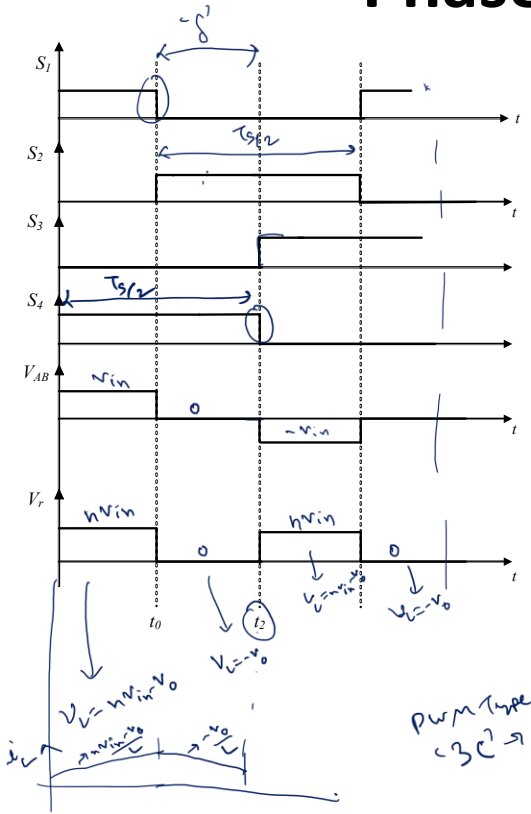
Recap

→ PWM strategies for FBC (9 possibilities)

PWM Type-1 (1,2,3) → diagonal switches turn-off at the same time

PWM Type-2 (4,5--9) → diagonal switches turn-off at different time

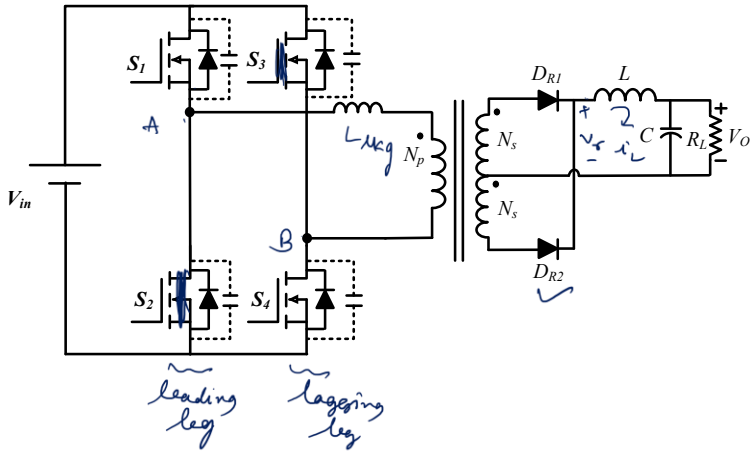
Phase-Shifted Full Bridge Converter



Assume ideal switch

$t_{on}(S_1, S_2, S_3, S_4) \rightarrow \frac{T_s}{2}$

$\delta = \frac{t_2 - t_0}{T_s/2} \cdot 180^\circ$



$n = \frac{N_s}{N_p}$

$\frac{\delta}{180^\circ} = \frac{t_2 - t_0}{T_s/2} = 1 - D$

$\Rightarrow \frac{\delta}{180^\circ} \cdot \frac{T_s}{2} = (1-D)\frac{T_s}{2}$

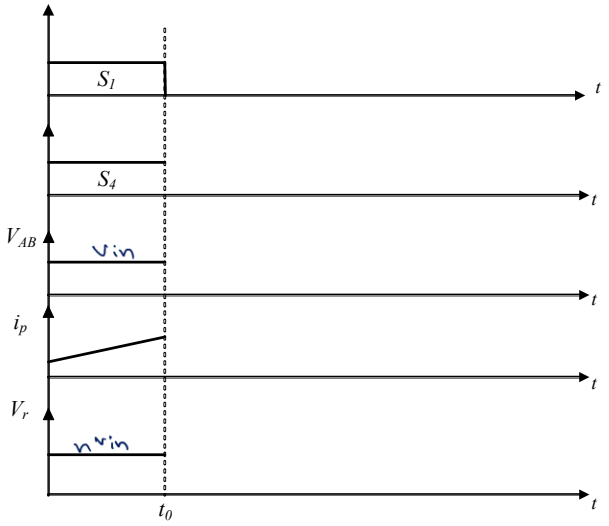
$\Rightarrow \delta \uparrow \Rightarrow$ O/p voltage $(v_o) \downarrow$

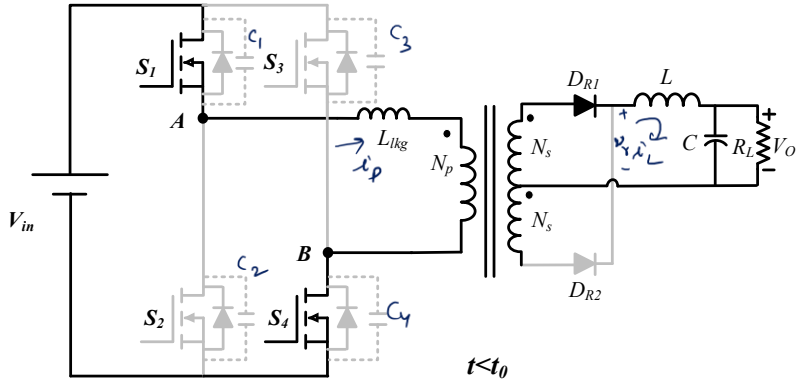
$(v_o = nV_{in}D)$

PWM Type-II
'3C' → Modulation strategy

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-0)





S_1 & S_4 is conducting

$v_{C4} = 0$

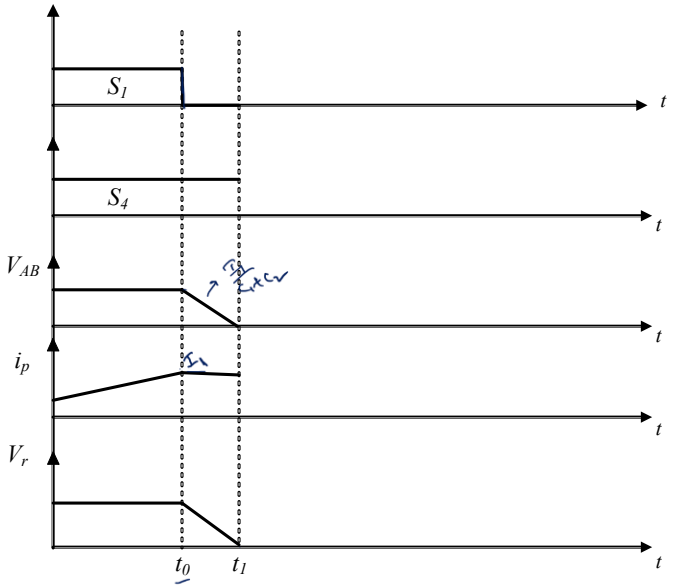
$v_{C3} = v_{in}$

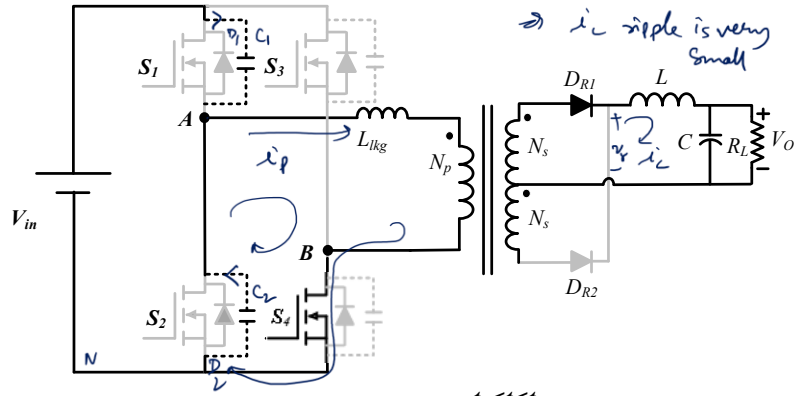
$v_{C1} = 0$

$v_{C2} = v_{in}$

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-1)





$t_0 < t < t_1$

L is large
⇒ i_L ripple is very small

C_2 is discharging from $V_{in} \rightarrow 0$
⇒ $v_{C_2} \downarrow$

C_1 is charging from 0
⇒ $v_{C_1} \uparrow$

$v_{AB} = v_{C_2}$

$i_{C_1} + i_{C_2} = i_p$

$\Rightarrow \frac{dv}{dt} = \frac{I_1}{C_1 + C_2}$

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-1) Cont.

$i_p(t_0) \simeq i_p(t_1) \simeq I_1$

$v_{c_1}(t) = \frac{I_1}{C_1 + C_2}(t - t_0)$ $\quad (C_1 \simeq C_2 \simeq C)$

$\Rightarrow v_{c_1}(t) = \frac{I_1}{2C}(t - t_0)$

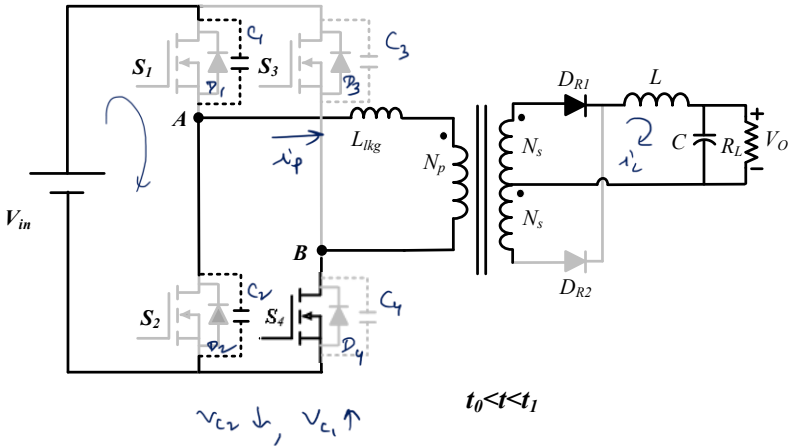
$\Rightarrow v_{c_2}(t) = V_{in} - \frac{I_1}{2C}(t - t_0)$

at $t = t_1 \Rightarrow v_{c_1}(t_1) = V_{in}$

$\Rightarrow v_{c_2}(t_1) = 0$

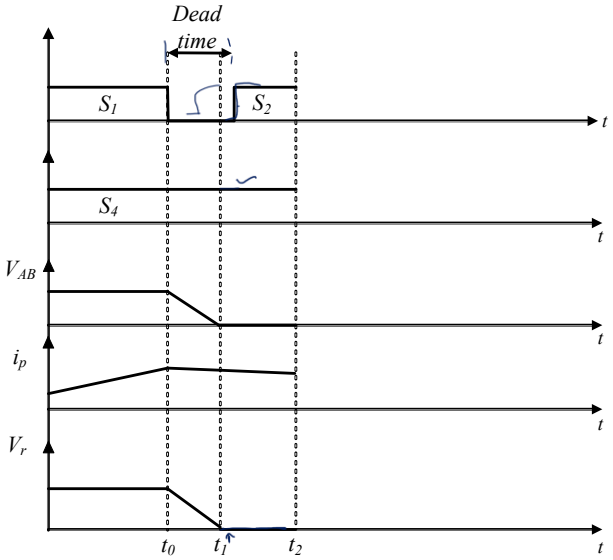
$\Rightarrow t_{01} = t_1 - t_0 = \frac{2C V_{in}}{I_1}$

after t_{01} → the $v_{c_2} = 0$ ⇒ the diode can turn-on

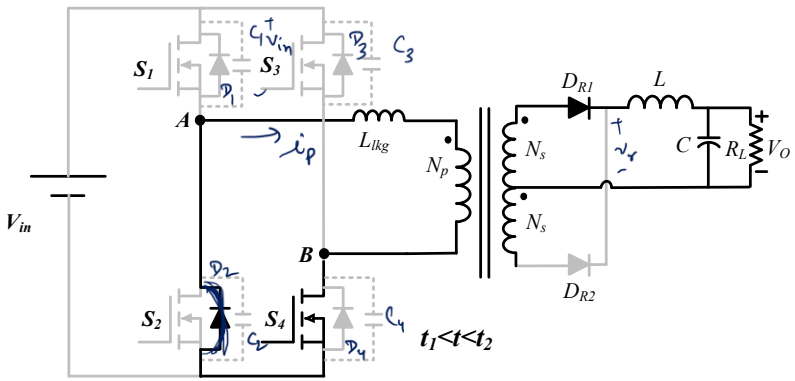


Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-2)



$V_{C_1} = V_{in}$



⇒ Since the $v_{C_2} = 0$

⇒ the diode D_2 turn-on

$i_{D_2} = i_p$

Since D_2 is on, after some time the switch S_2 can turn-on with ZVS

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-2)

if, Deadtime $> t_{01}$ ⟶

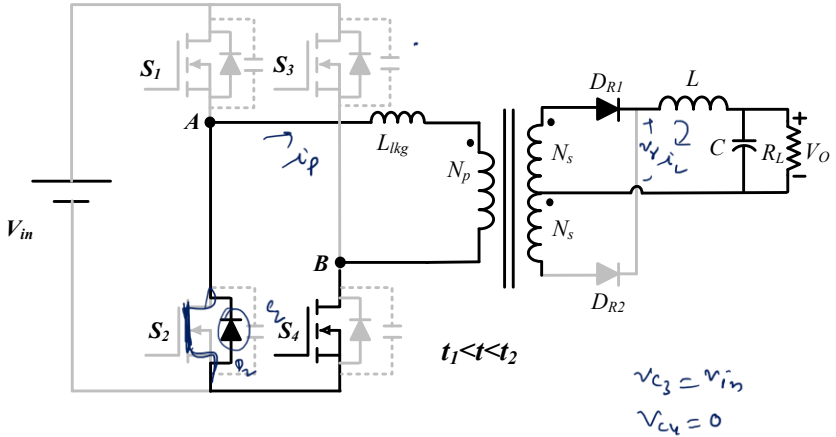
$\simeq$ Deadtime $> \frac{2CV_{in}}{I_1}$ $\Rightarrow$ ZVS turn on of S_2 switch is possible

$\Rightarrow$ after some time between t_1 & t_2 the gate pulse to S_2 can be given, which ensure ZVS turn-on of S_2 switch (leading leg switch)

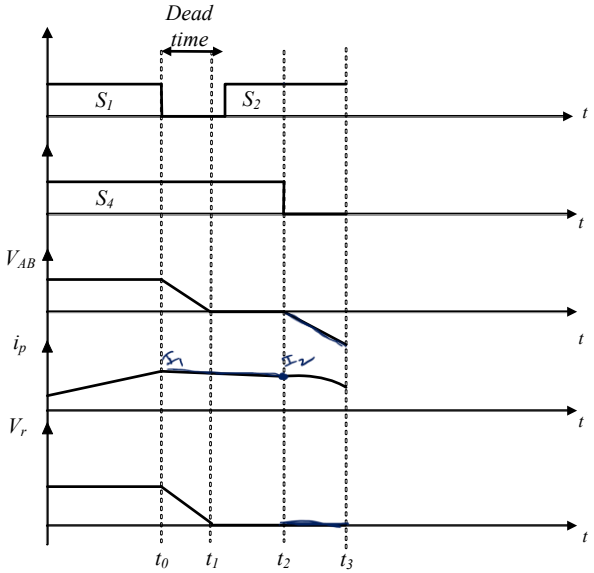
to ensure ZVS, $\Rightarrow$ Deadtime (leading leg) $> \frac{2CV_{in}}{I_1}$ $\left(C \simeq C_1 \simeq C_2\right)$

$i_L \simeq I_o$

$\Rightarrow$ $i_p \simeq n i_L$ $\Rightarrow$ $I_1 \simeq n I_o$

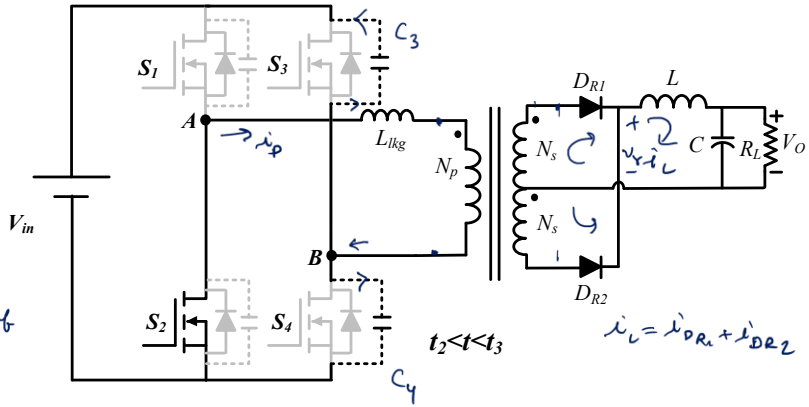


PSFB Operating Modes (Mode-3)



$t_2 - t_0$ = Phase-shift

at $t = t_2$ the Switch S_4 → turns-off (lagging leg Switch)



C_3 is discharging from V_{in}

C_4 is charging from 0V

⇒ $V_{AB} = -V_{C_4}$ (V_{C_4} ↑)

⇒ i_p will no longer be reflected i_L current

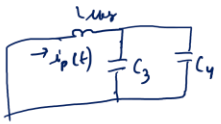
⇒ D_{R_1} & D_{R_2} gets forward biased ⇒ $v_r = 0$

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-3)

V_{AB} is imposed upon L_{lkg}

L_{lkg} is in resonance with $(C_3 + C_4)$



$$i_p(t) = I_2 \cos(\omega_1 (t - t_2))$$

$$v_{C_4}(t) = \frac{1}{2C} \int i_p(t) \cdot dt$$

$$= \frac{1}{2C\omega_1} I_2 \sin(\omega_1 (t - t_2))$$

$$v_{C_4}(t) = \sqrt{\frac{L_{lkg}}{2C}} \cdot I_2 \sin(\omega_1 (t - t_2))$$

$$\Rightarrow v_{C_3}(t) = v_{in} - v_{C_4}(t)$$

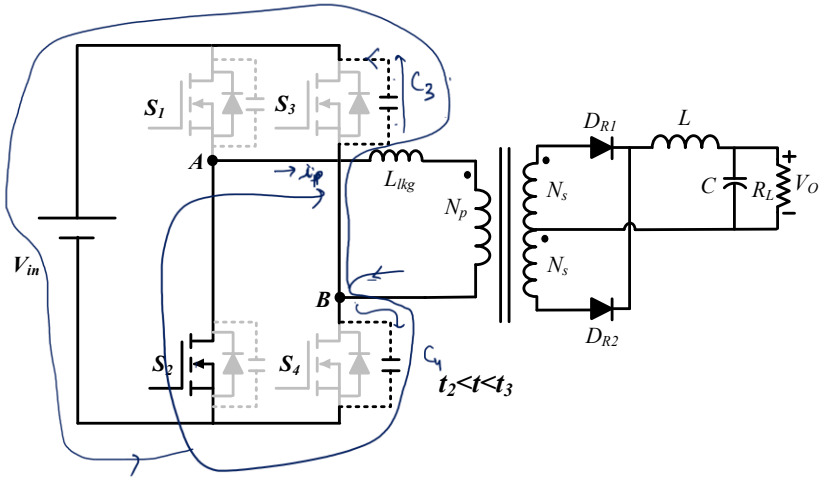
$$= v_{in} - \sqrt{\frac{L_{lkg}}{2C}} \cdot I_2 \sin(\omega_1 (t - t_2))$$

ω_1 → resonance freq.

$$\omega_1 = \frac{1}{\sqrt{L_{lkg}(C_3 + C_4)}}$$

$$\Rightarrow C_3 \approx C_4 = C$$

$$\Rightarrow \omega_1 = \frac{1}{\sqrt{2 L_{lkg} C}}$$



Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

Thank You